Module 2: Enhanced ER Modeling

COMP 2714 – Relational Database Systems

Learning Outcome: LO2 – Apply techniques of database design including enhanced entity–relationship modeling.

Blueprint for a database?

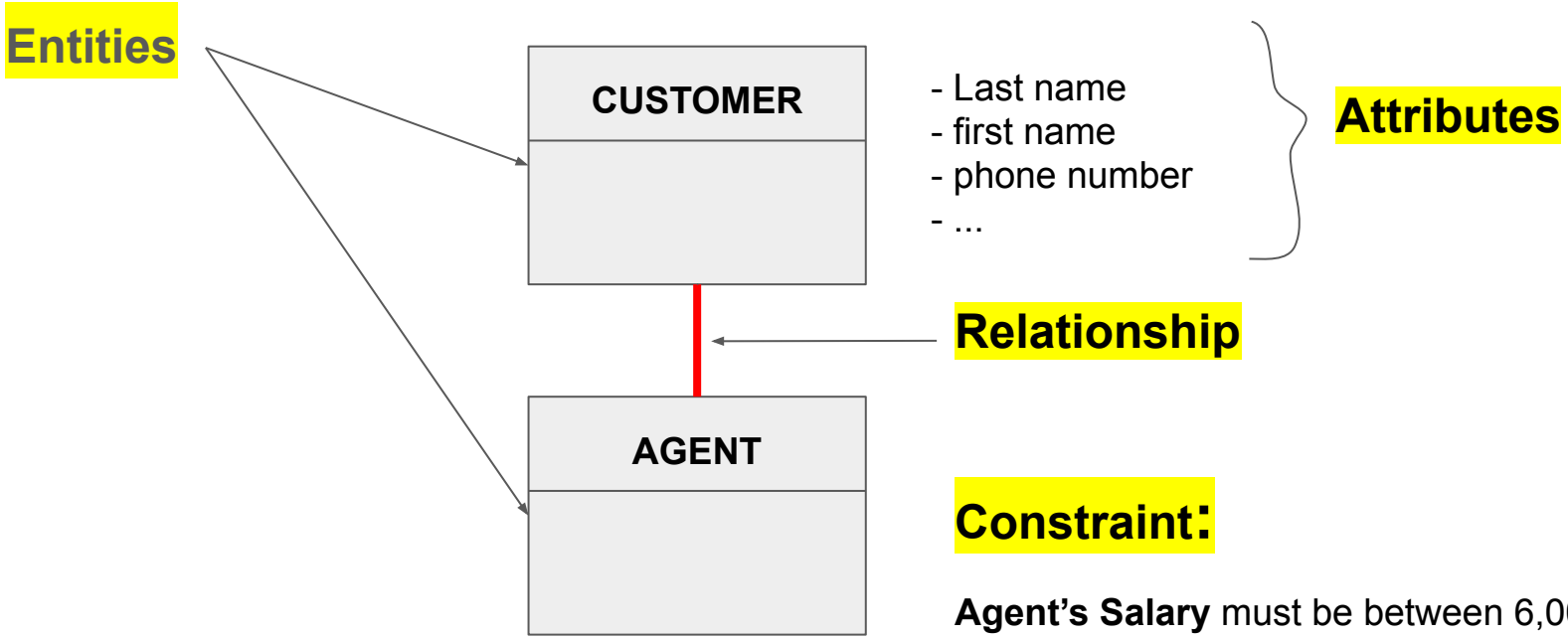
- Like building **cars** or building **houses**, building a database needs a blueprint
- We need to create a **model first** and then implement it

How do we create a model for a database?

What is a data model?

- Remember that a vital part of the database designer's job is to determine **what data** to enter into the database and **how to organize** that data
- A data model defines and describes the **structures** for storing data , the **rules** for ensuring data integrity and the methodologies for data **access** and manipulation
- A data model is both **narrative & graphical**

Data Model Building Blocks



Constraint:

Agent's Salary must be between 6,000 - 110,000

Business Rules

To **discover** business rules ask these questions:

- What data exists?
- How is it used?

You should **write down** the business rules as you discover them and try to verify their validity by asking multiple people in an organization to share their view of data and what they are using it for.

Business rules **establish entities and the relationships** between them as well as data **constraints**.

Examples of Business Rules

- **A pilot** cannot be on duty for more than 10 hours during a 24-hour period.
- **A professor** may teach up to four classes during a semester.

Entity Relationship Diagram (ERD)

Entity

- A person, place, thing, concept, or event for which data can be stored.
- written in CAPITAL LETTERS

Relationship

- An association between entities.
- Ask two questions **to identify the type** of each relationship:
    - How many CUSTOMERs does an AGENT serve?
    - How many AGENTs serve a single CUSTOMER?

Relationship Types

One to Many → 1:M or 1..*

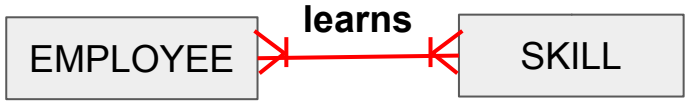
- A painter paints many paintings
- Each painting is painted by only one painter

One to One → 1:1 or 1..1

- Each store is managed by a single employee
- Each store manager manages only a single store

Many to Many→ M:N or *..*

- An employee may learn many job skills
- Each job skill may be learned by many employees



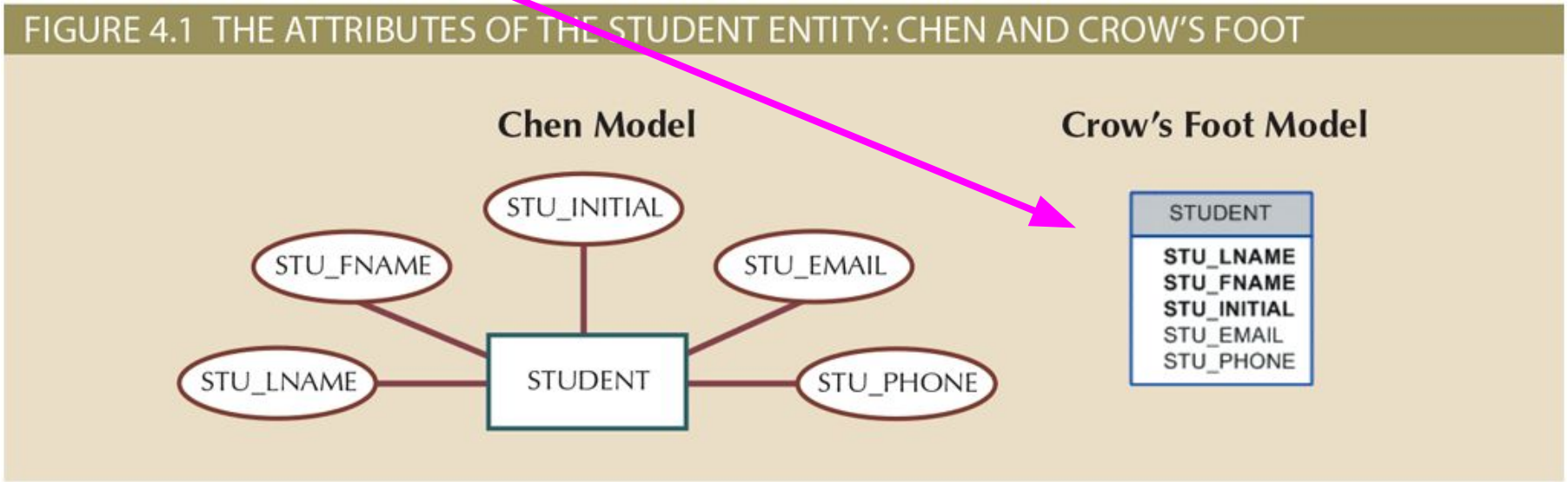
ER Modeling

A **deeper dive** into building a blueprint for a relational DB

1. Characteristics of **Attributes**

Required or Optional?

- **Required** or **Optional**
- **Bold-faced** in CF notation



PKs

- Every entity must have an identifier
- **Identifiers (PKs)** are underlined in ERDs
  - Pk is ideally one attribute
  - But sometimes a composite

Simple or Composite?

Attributes are either **simple** or **composite**

Can be divided to yield additional attributes
(e.g. ADDRESS)

Always be in the lookout for composite attributes

→ Business rules talk often in terms of composite attributes

→ Designer should recognize them + determine the correct way to **decompose**

Single-Valued or Multivalued?

Attributes can be **single-valued** or **multivalued**

- Can have only a single value, e.g.
  - SIN for person
  - Serial# of manufactured parts
- Single-valued is **not necessarily simple**
  - SE-08-02-189935

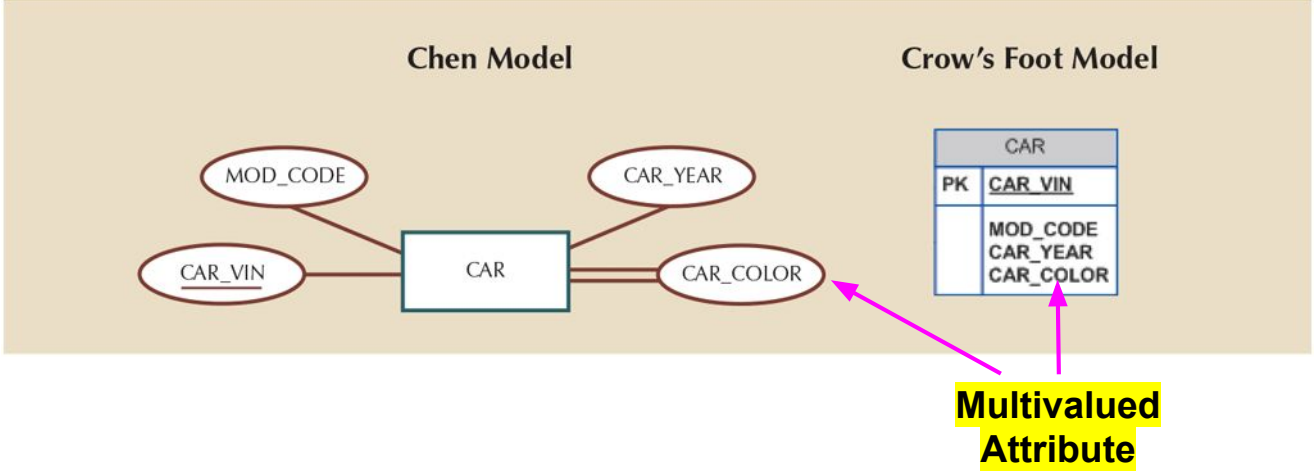
Region Plant Shift Part #

- Can have many values, e.g.
  - Person → several degrees
  - Household → several phones
  - Car's color -> roof / trim / body

Modeling Multivalued Attributes

- in **ER modeling**, attributes can be **multi-valued**
- In the **relational model**, **multi-valued attributes are not allowed**
- They must be **transformed** before implementation
  - (Details → Module 3: Mapping ER → Relational Schema)
- For now: just **recognize them in ERDs** and know they will require special handling

FIGURE 4.3 A MULTIVALUED ATTRIBUTE IN AN ENTITY

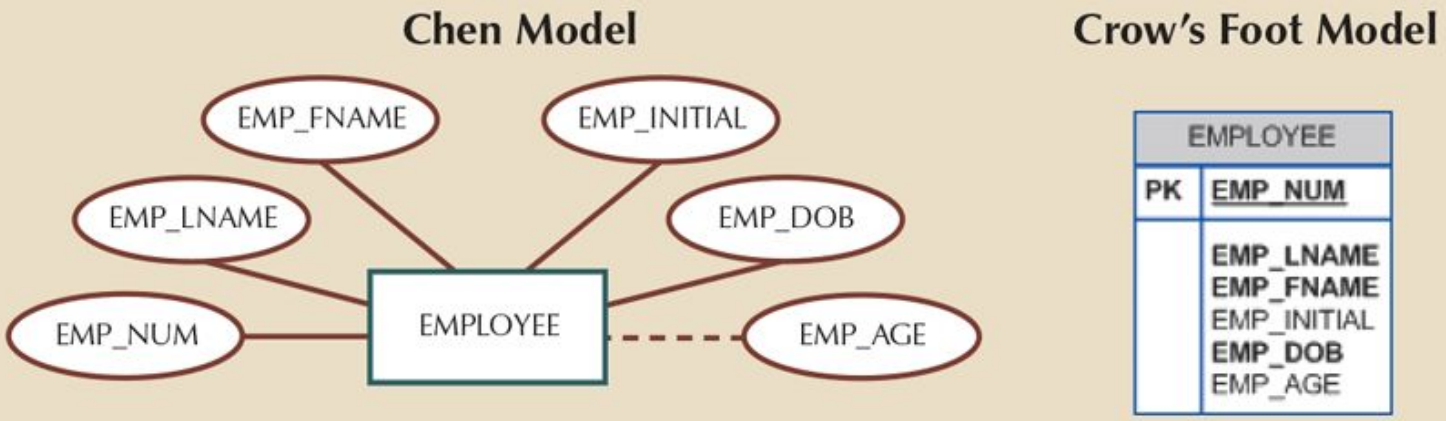


Derived (Computed) Attributes

The value of the attribute is **derived from other attributes**

e.g. Age, Total Cost of Orders, Avg Speed

FIGURE 4.6 DEPICTION OF A DERIVED ATTRIBUTE



Implementing Derived Attributes

Designer should be aware of **advantages & disadvantages** of storing derived attributes

TABLE 4.2

ADVANTAGES AND DISADVANTAGES OF STORING DERIVED ATTRIBUTES

	DERIVED ATTRIBUTE	
	STORED	NOT STORED
Advantage	Saves CPU processing cycles Saves data access time Data value is readily available Can be used to keep track of historical data	Saves storage space Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	Uses CPU processing cycles Increases data access time Adds coding complexity to queries

Attributes Dive In Checklist

- Specify **required** attributes
- Specify **identifiers** (PKs)
- Are there any **composite attributes**?
  - Do we need to decompose?
- Specify **multivalued** attributes
  - What is the best way to implement them?
- Should we define any **derived attributes**?

2. Characteristics of **Relationships**

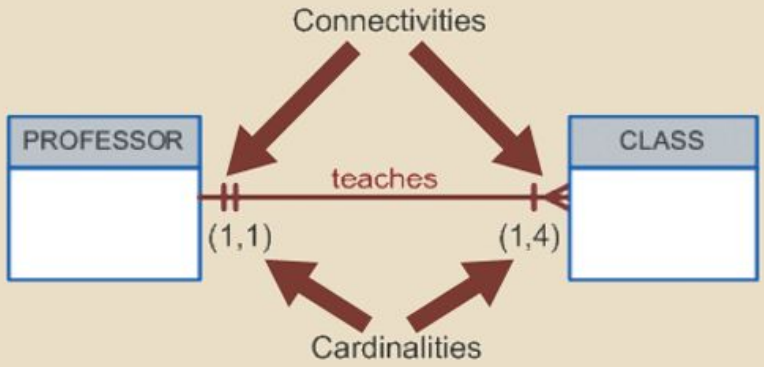
Relationships

- **Participants:** entities in relationships
- **Name** of the relationship → Active or passive verb
- Always **bidirectional** → you should know both sides to be able to detect the type of the relationship

Connectivity & Cardinality

- **Connectivity**: Relationship Classification (1:1, 1:M, M:N)
- **Cardinality**: Expresses **min & max** number of instances in relationship
  - Cardinality cannot be implemented in the DBMS directly
  - Application software or triggers can handle cardinality

FIGURE 4.7 CONNECTIVITY AND CARDINALITY IN AN ERD



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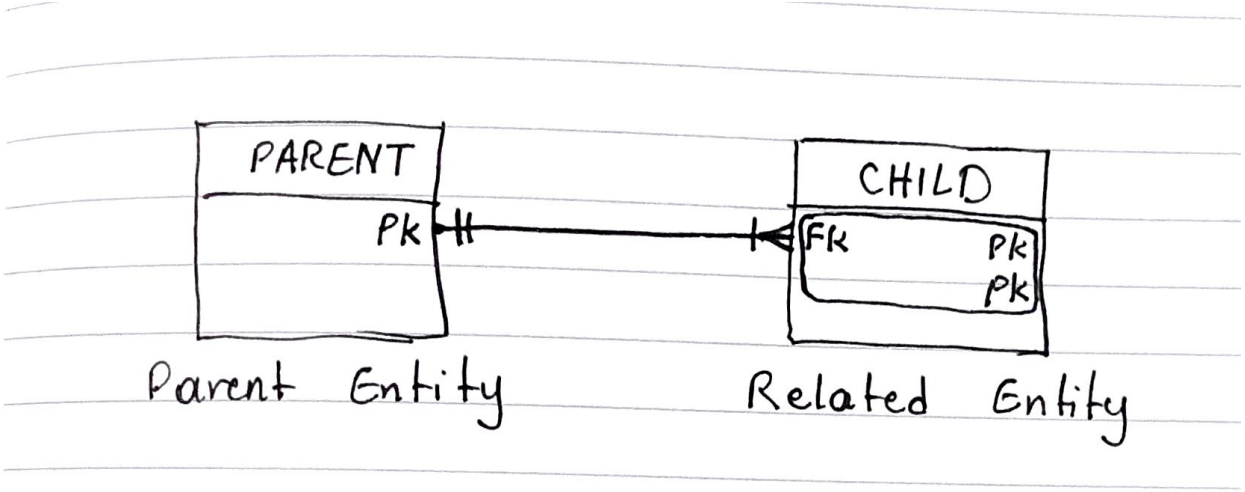
Existence Dependence

- Some entities, we should keep in the database, only if they are related to another entity in our system
- Example: “**EMPLOYEE Claims DEPENDENT**”
- If the employee is removed from the database, we clearly should remove his dependents from the database as well.

3. Relationship **Strength**

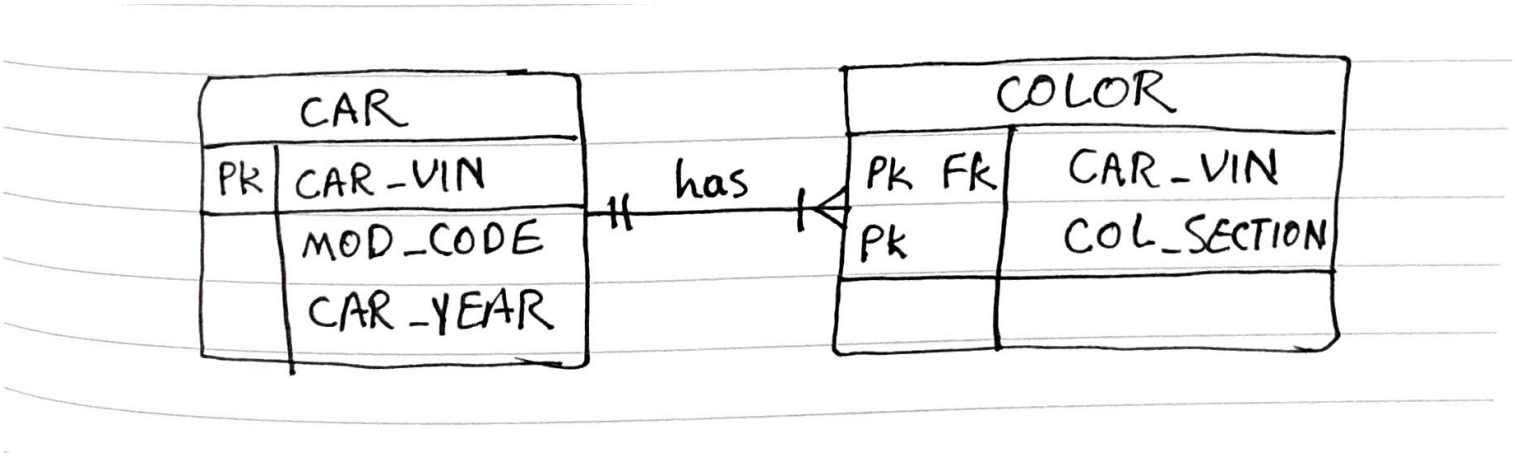
Or permanence over time

Strong Relationships



Sometimes the primary key of the child entity includes the primary key of the parent. This kind of parent-child relationships are called **strong relationships.**

Modeling Strong Relationships



Strong relationships (also called **identifying relationships**) are specified with **solid lines** in Crow's Foot ERDs.

FIGURE 4.8 A WEAK (NON-IDENTIFYING) RELATIONSHIP BETWEEN COURSE AND CLASS

Table name: COURSE

Database name: Ch04_TinyCollege

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	W 6:00-8:40 p.m.	KLR209	162
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

Weak Relationships

The PK of the child **does not inherit** a component from the parent PK

The parent's PK is only a foreign key in the child table (and not part of child's primary key)

Weak relationships are specified by **dashed lines** in Crow's Foot ERDs.

Weak Entities

!DO NOT CONFUSE WITH WEAK RELATIONSHIPS!

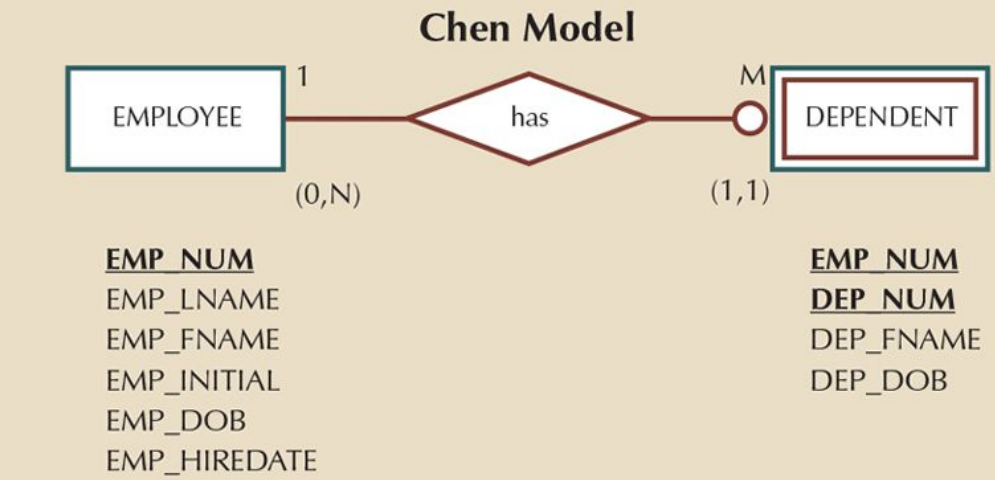
Two Conditions - Both are required for an entity to be weak:

1. Existence Dependent → Entity **cannot exist without the parent** entity
2. Has a Primary Key that is partially or totally derived from the parent key (**Strong relationship with the parent**)

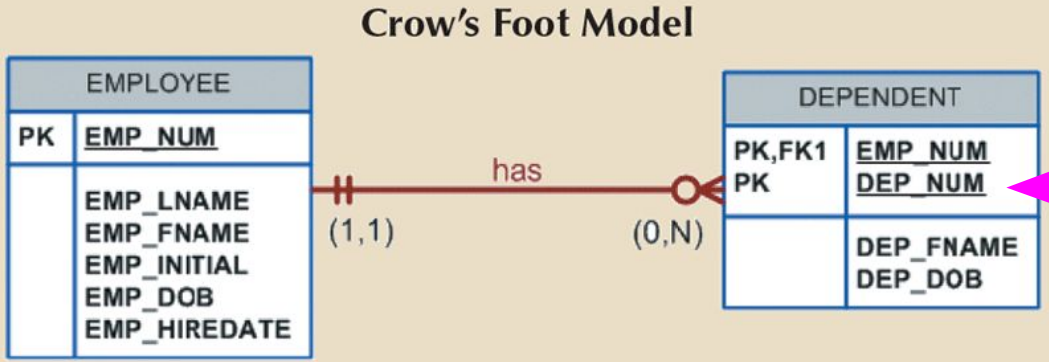
In other words, an entity can be existence-dependent but not weak

Modeling Weak Entities

FIGURE 4.10 A WEAK ENTITY IN AN ERD



Double walled → Weak Entity



In Crow's Foot, weak entities are recognizable by noticing the **strong relationship they have with parent entity** and the fact that the child's PK includes parent's PK

Associative (Composite) Entities

FIGURE 4.23 CONVERTING THE M:N RELATIONSHIP INTO TWO 1:M RELATIONSHIPS

Table name: STUDENT **Database name: Ch04_CollegeTry**

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Table name: ENROLL

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114

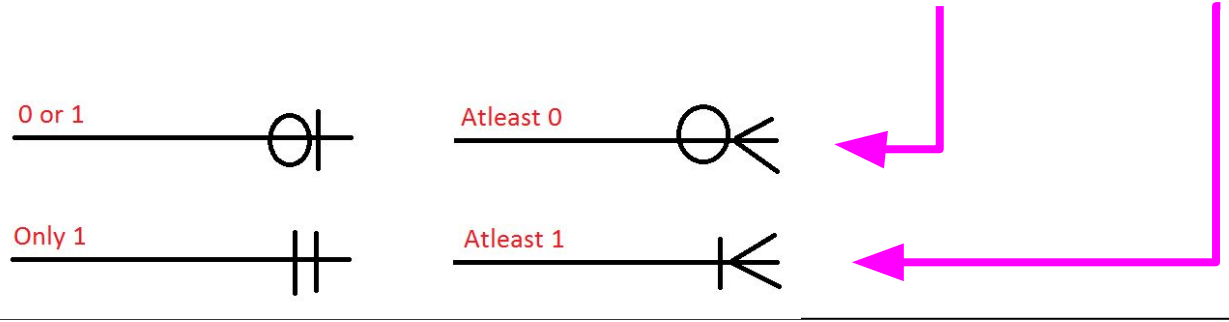
- ENROLL is **existence-dependent** on the two other entities

PK: {CLASS_CODE, STU_NUM}

Associative/Composite/Bridge entities are often **modeled as weak entities.**

Relationship Participation

An entity's participation in a relationship might be **optional** or **mandatory**



Relationship Participation ≠ Relationship Strength

- Possible to have strong but optional relationships
  - e.g. Employee claims Dependent
- Possible to have weak but mandatory relationships

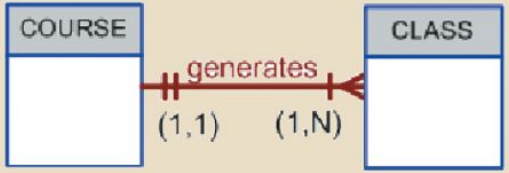
Relationship Participation

- Class is **optional to** professor
  - professors who conduct research without teaching classes
- Professor is **mandatory to** class

FIGURE 4.12 AN OPTIONAL CLASS ENTITY IN THE RELATIONSHIP "PROFESSOR TEACHES CLASS"

FIGURE 4.13 CLASS IS OPTIONAL TO COURSE

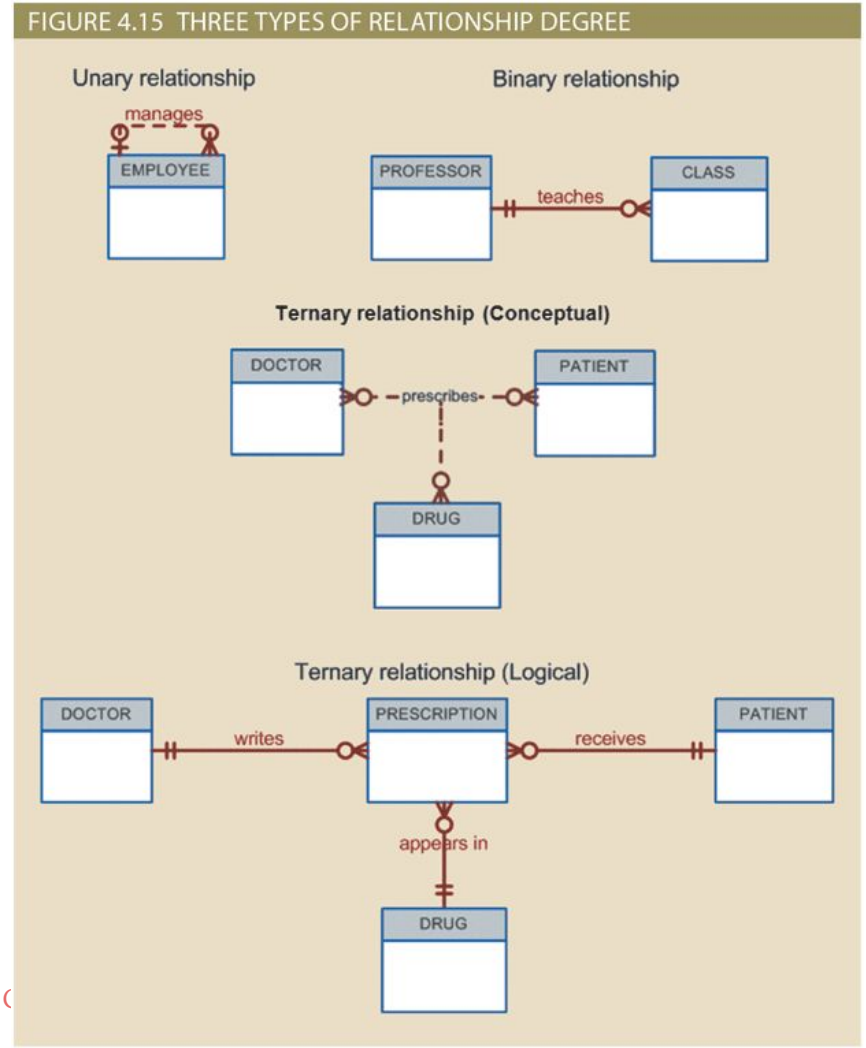
FIGURE 4.14 COURSE AND CLASS IN A MANDATORY RELATIONSHIP



Relationship Participation

COURSE is mandatory in the relationship. But the participation of the CLASS entity might be **optional or mandatory**.

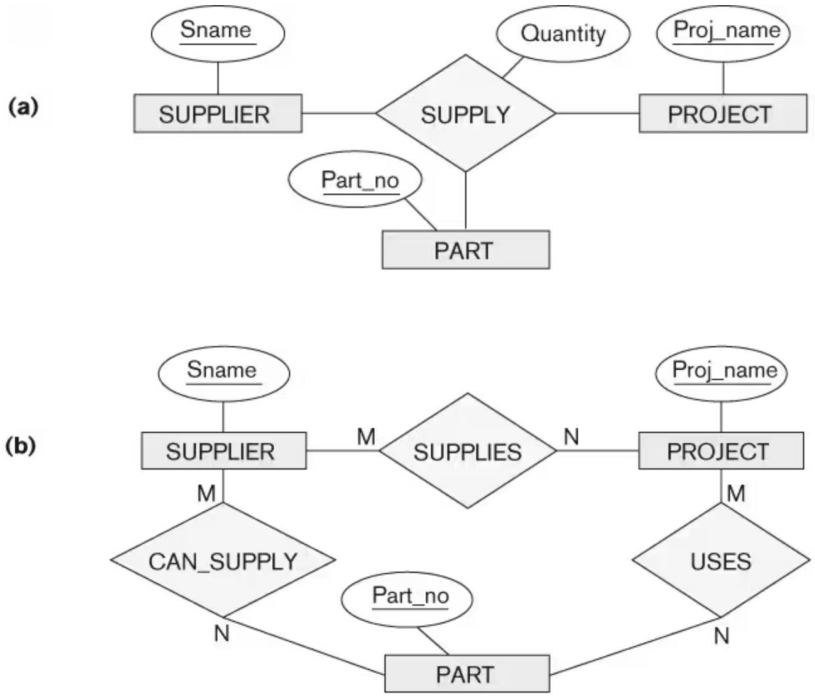
Optionality Reflects Practice



Relationship Degree

- Unary
- Binary
- Ternary

Ternary Relationships

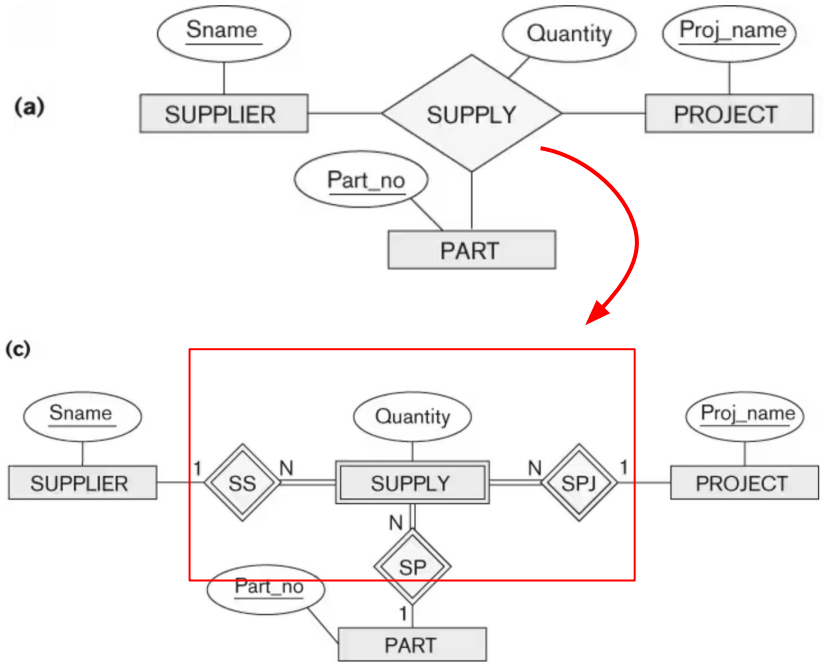


The SUPPLY relationship is a ternary relationship. And even though there are 3 binary relationships between:

- SUPPLIER & PROJECT
- SUPPLIER & PART
- PROJECT & PART

These 3 relationships together are not equivalent to the ternary relationship.

Ternary Relationships



Ternary relationships are **modelled as weak entities.**

PK: {Sname, Proj_name, Part_no}

Recursive Relationships

- Naturally occurs **in unary relationships**
- A recursive relationship might be **1:1, 1:M or M:N**

FIGURE 4.17 AN ER REPRESENTATION OF RECURSIVE RELATIONSHIPS

EERD

- EERD builds on ER and adds support for entity **supertypes**, **subtypes** and **clusters**.
- Sometimes referred to as **Enhanced** ER Model.
- EERM adds **more semantic constructs** to the original ER Model.

Supertypes & Subtypes

- The following is example data for **all employees** hired by an aviation business: pilots, mechanics, accountants, DB designers, etc.

FIGURE 5.1 NULLS CREATED BY UNIQUE ATTRIBUTES

Database name: Ch05_AirCo

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_LICENSE	EMP_RATINGS	EMP_MED_TYPE	EMP_HIRE_DATE
100	Kolmycz	Xavier	T				15-Mar-88
101	Lewis	Marcos		ATP	SEL/MEL/Instr/CFII	1	25-Apr-89
102	Vandam	Jean					20-Dec-93
103	Jones	Victoria	R				28-Aug-03
104	Lange	Edith		ATP	SEL/MEL/Instr	1	20-Oct-97
105	Williams	Gabriel	U	COM	SEL/MEL/Instr/CFI	2	08-Nov-97
106	Duzak	Mario		COM	SEL/MEL/Instr	2	05-Jan-04
107	Diante	Venite	L				02-Jul-97
108	Wiesenbach	Joni					18-Nov-95
109	Travis	Brett	T	COM	SEL/MEL/SES/Instr/CFII	1	14-Apr-01
110	Genkazi	Stan					01-Dec-03

Supertypes & Subtypes

- The following is example data for **all employees** hired an aviation business: pilots, mechanics, accountants, DB designers, etc.
- But these attributes only apply to pilots not accountant or mechanics.

FIGURE 5.1 NULLS CREATED BY UNIQUE ATTRIBUTES

Database name: Ch05_AirCo

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_LICENSE	EMP_RATINGS	EMP_MED_TYPE	EMP_HIRE_DATE
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102	Vandam	Jean					20-Dec-93
103	Jones	Victoria	R				28-Aug-03
104	Lange	Edith		ATP	SEL/MEL/Instr	1	20-Oct-97
105	Williams	Gabriel	U	COM	SEL/MEL/Instr/CFI	2	08-Nov-97
106	Duzak	Mario		COM	SEL/MEL/Instr	2	05-Jan-04
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109	Travis	Brett	T	COM	SEL/MEL/SES/Instr/CFII	1	14-Apr-01
110	Genkazi	Stan					01-Dec-03

Supertypes & Subtypes

- When some employees have characteristics that are not shared by other employees, we see **unnecessary nulls** in the table.

FIGURE 5.1 NULLS CREATED BY UNIQUE ATTRIBUTES

Database name: Ch05_AirCo

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_LICENSE	EMP_RATINGS	EMP_MED_TYPE	EMP_HIRE_DATE
100	Kolmycz	Xavier	T				15-Mar-88
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106	Duzak	Mario		COM	SEL/MEL/Instr	2	05-Jan-04
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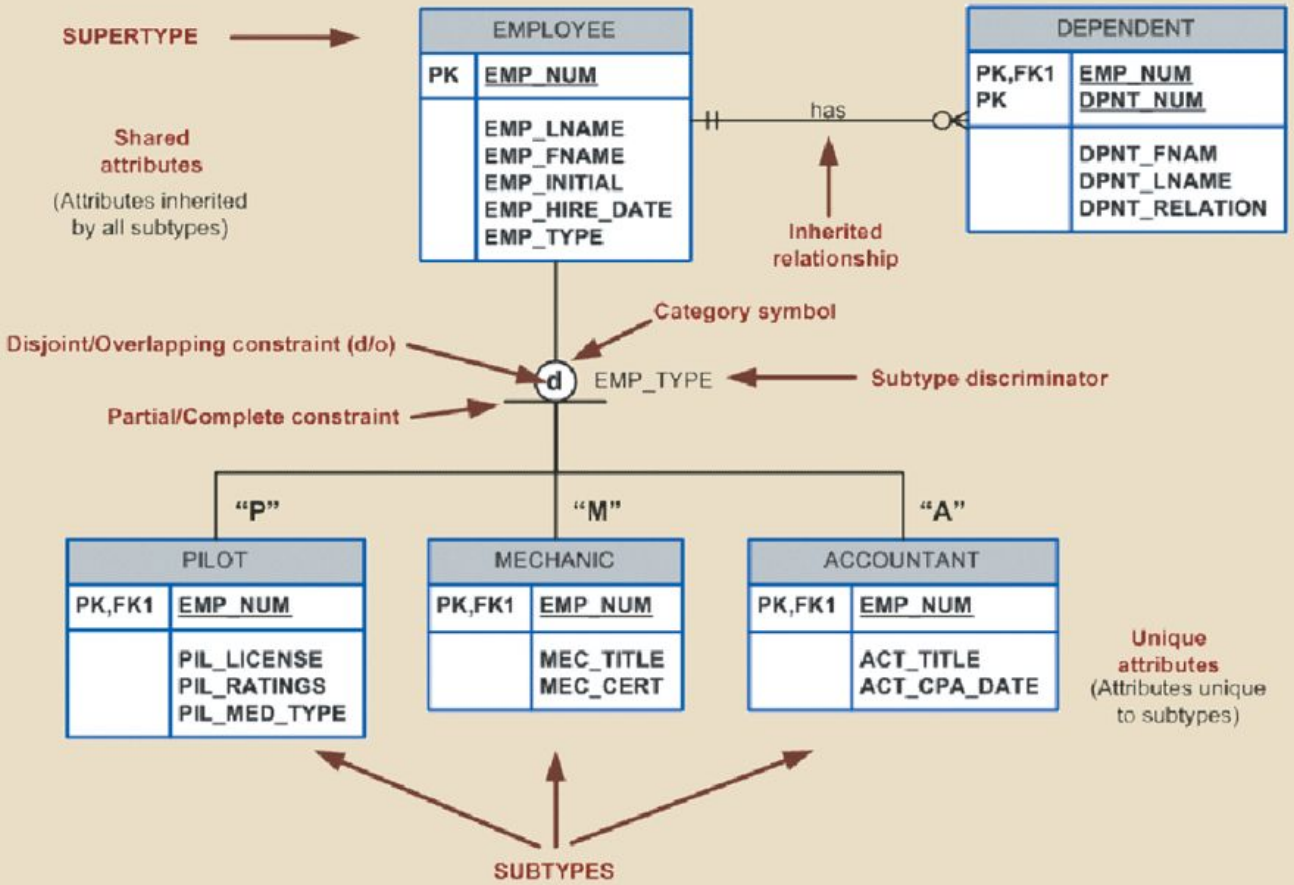
108	Wiesenbach	Joni					18-Nov-95
109	Travis	Brett	T	COM	SEL/MEL/SES/Instr/CFII	1	14-Apr-01
110	Genkazi	Stan					01-Dec-03

FIGURE 5.2 A SPECIALIZATION HIERARCHY



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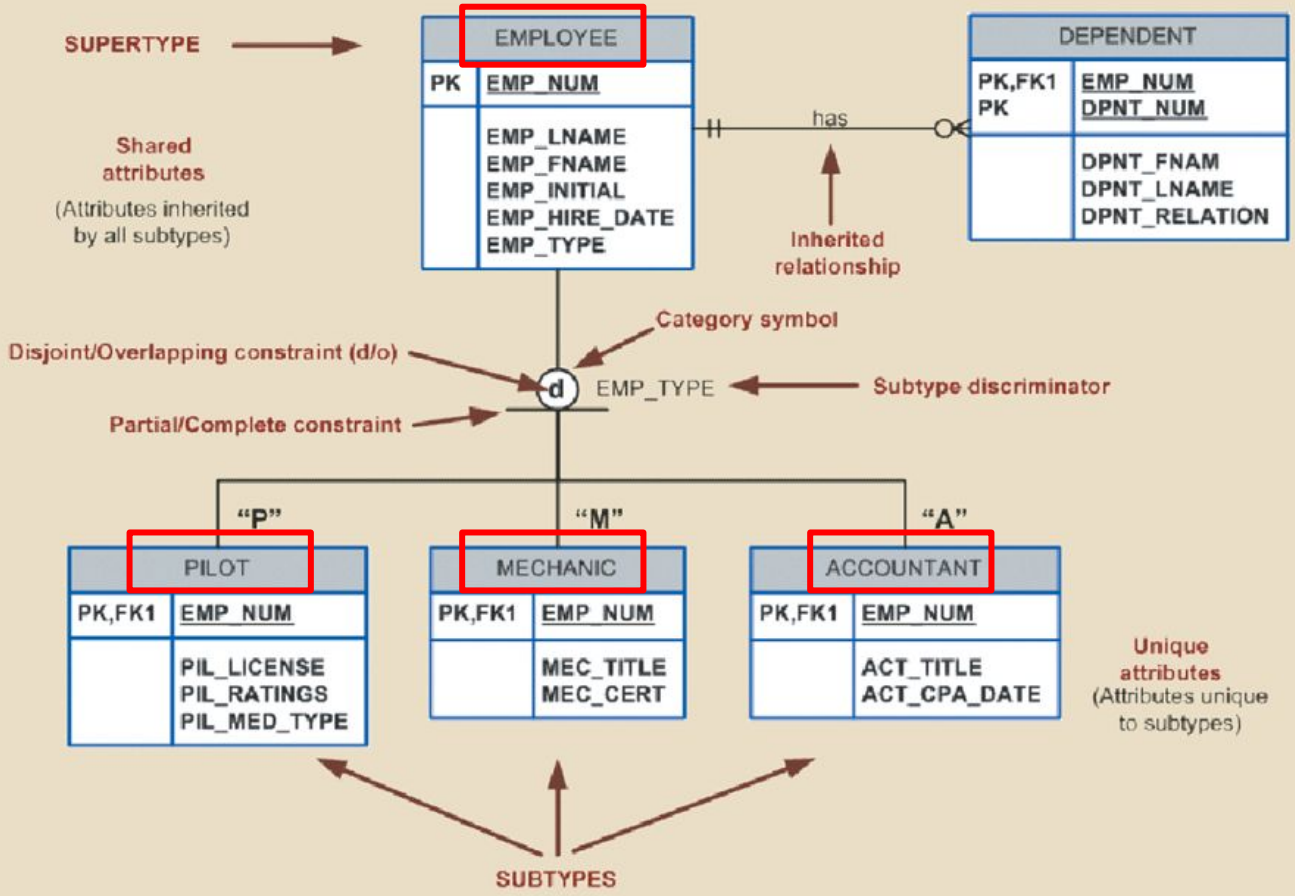
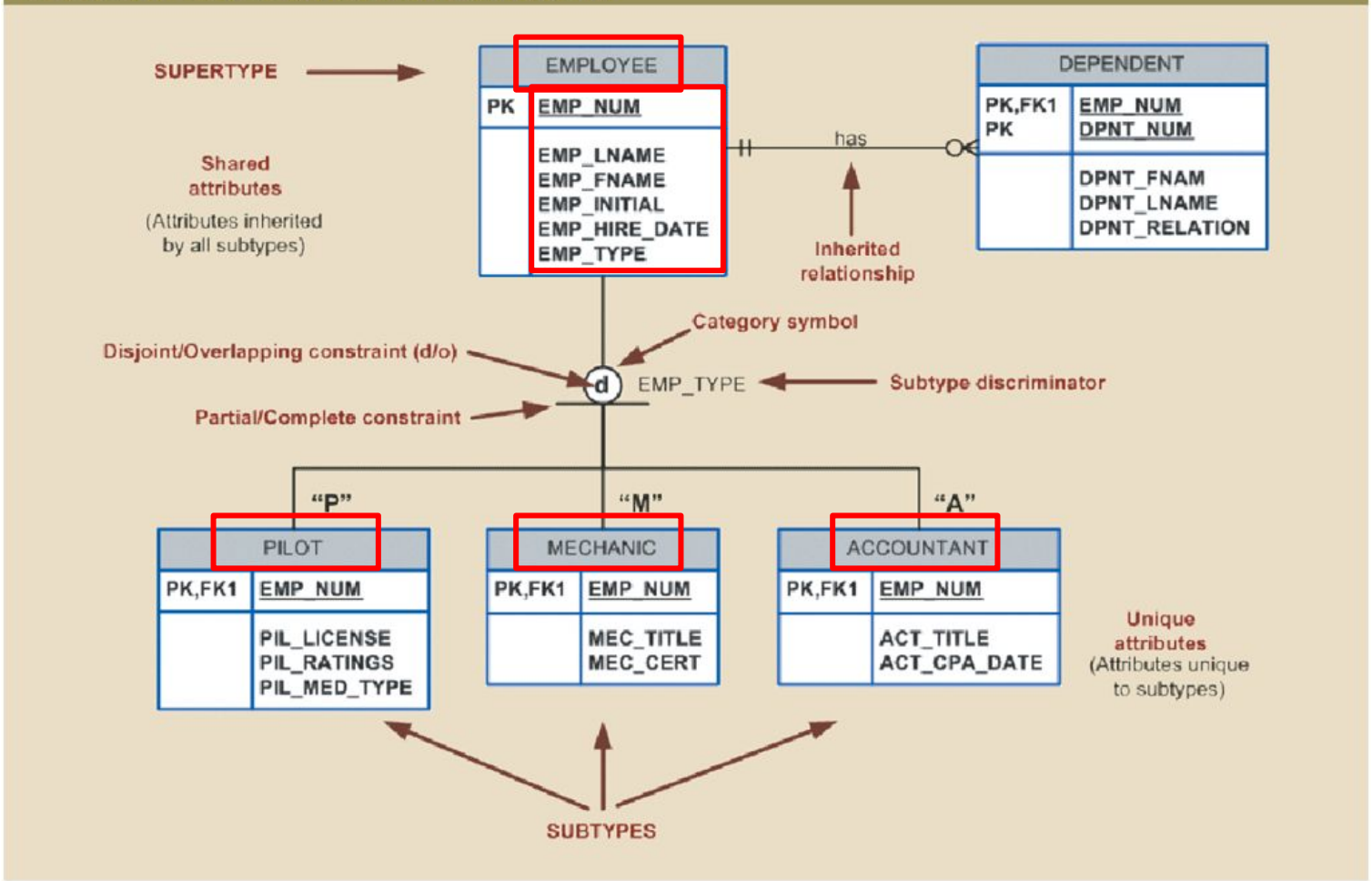


Supertype: a **generic** entity type that is related to one or more **subtypes**

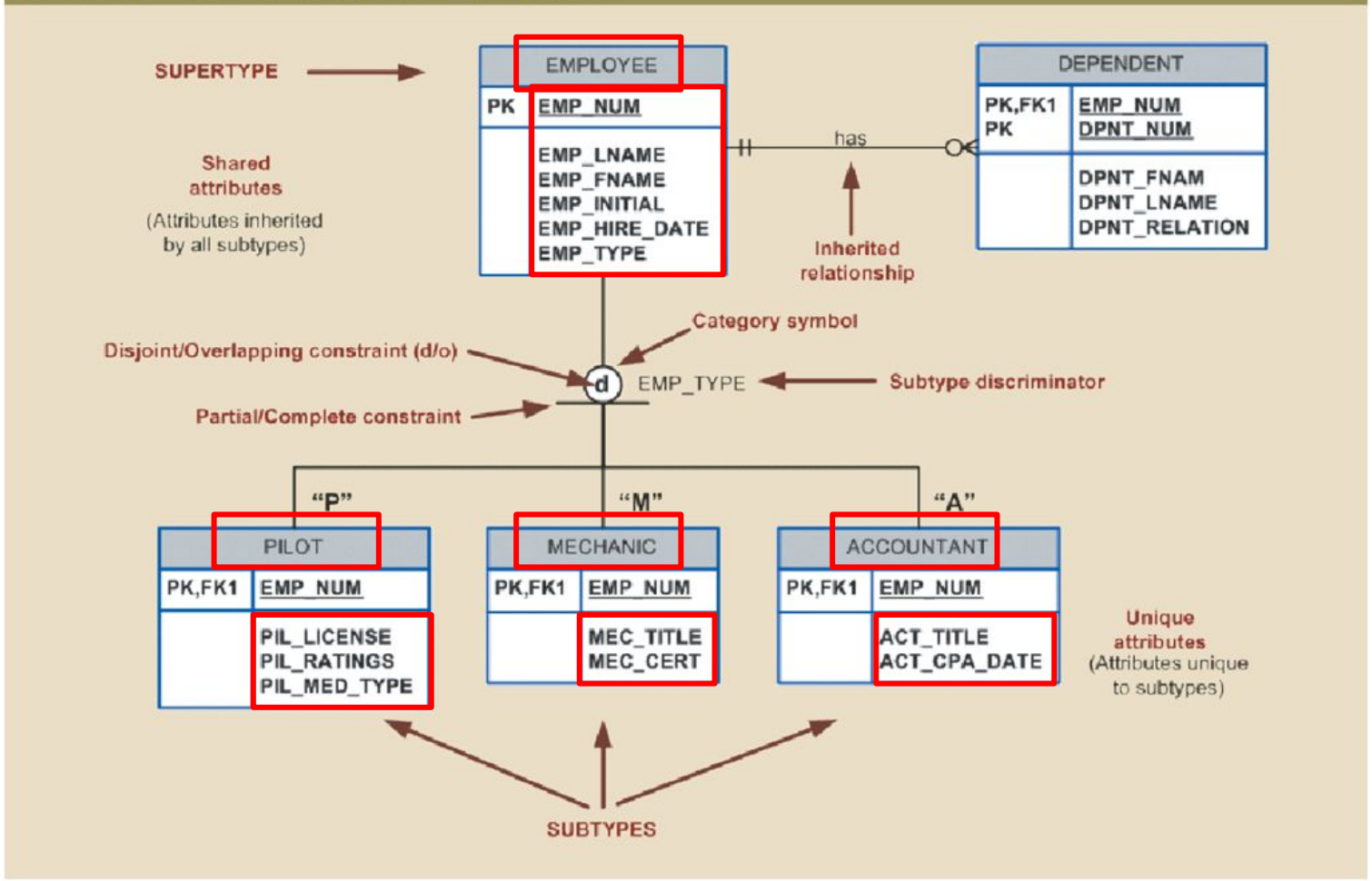
FIGURE 5.2 A SPECIALIZATION HIERARCHY



Supertype: a **generic** entity type that is related to one or more **subtypes**

supertype contains **common characteristics**

FIGURE 5.2 A SPECIALIZATION HIERARCHY



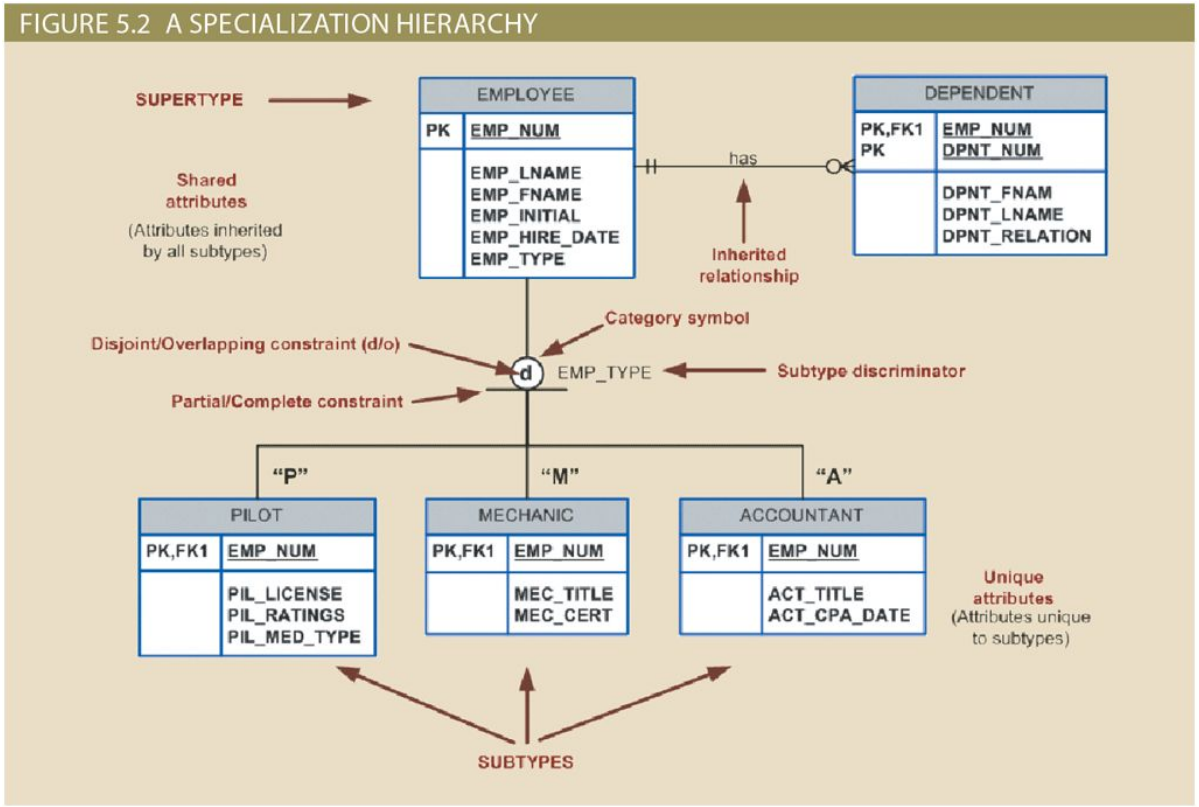
Supertype: a **generic** entity type that is related to one or more **subtypes**

supertype contains **common characteristics**

subtypes each contain their own **unique characteristics**

Use subtypes/supertypes **when** there are different kinds/types of the entity in the user environment ***AND*** each type has one or more attributes that is unique to its kind.

Specialization Hierarchy



A specialization hierarchy enable the data model to capture **additional semantic content**:

- attribute **inheritance**
- a special supertype attribute known as **subtype discriminator**
- **disjoint or overlapping** constraints
- **complete or partial** constraints

Inheritance

- the entity subtype inherits the **attributes and relationships** of the supertype
- all subtypes inherit their **primary key** attribute from their supertype
- Subtype and supertype have a **1:1 relationship** in implementation level

FIGURE 5.3 THE EMPLOYEE-PILOT SUPERTYPE-SUBTYPE RELATIONSHIP

Database name: Ch05_AirCo

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIRE_DATE	EMP_TYPE
100	Kolmycz	Xavier	T	15-Mar-88	
101	Lewis	Marcos		25-Apr-89	P
102	Vandam	Jean		20-Dec-93	A
103	Jones	Victoria	R	28-Aug-03	
104	Lange	Edith		20-Oct-97	P
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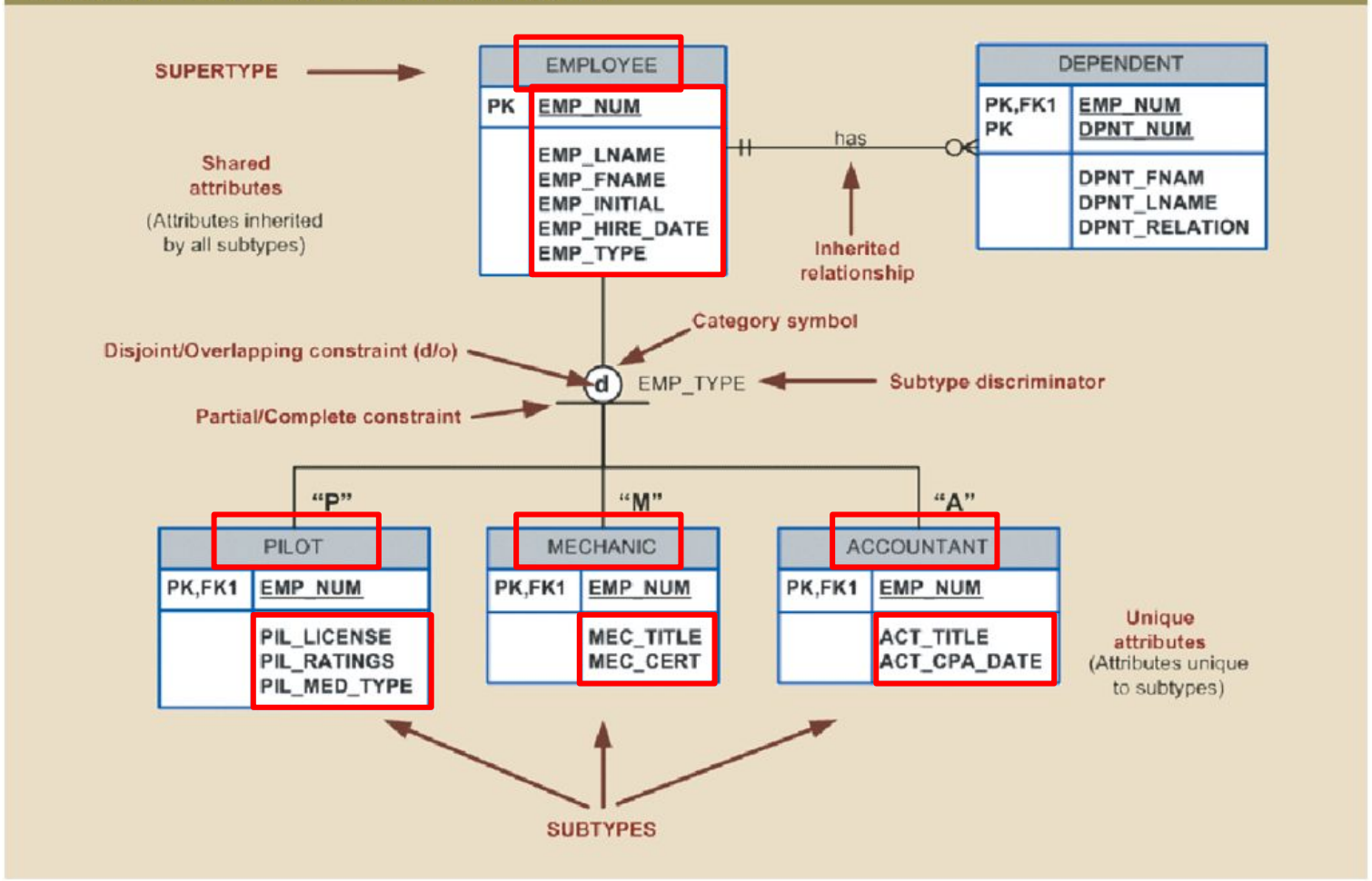
106	Duzak	Mario		05-Jan-04	P
107	Diante	Venite	L	02-Jul-97	M
108	Wiesenbach	Joni		18-Nov-95	M
109	Travis	Brett	T	14-Apr-01	P
110	Genkazi	Stan		01-Dec-03	A

Table name: PILOT

EMP_NUM	PIL_LICENSE	PIL_RATINGS	PIL_MED_TYPE
101	ATP	SEL/MEL/Instr/CFII	1
104	ATP	SEL/MEL/Instr	1
105	COM	SEL/MEL/Instr/CFI	2
106	COM	SEL/MEL/Instr	2
109	COM	SEL/MEL/SES/Instr/CFII	1

FIGURE 5.2 A SPECIALIZATION HIERARCHY

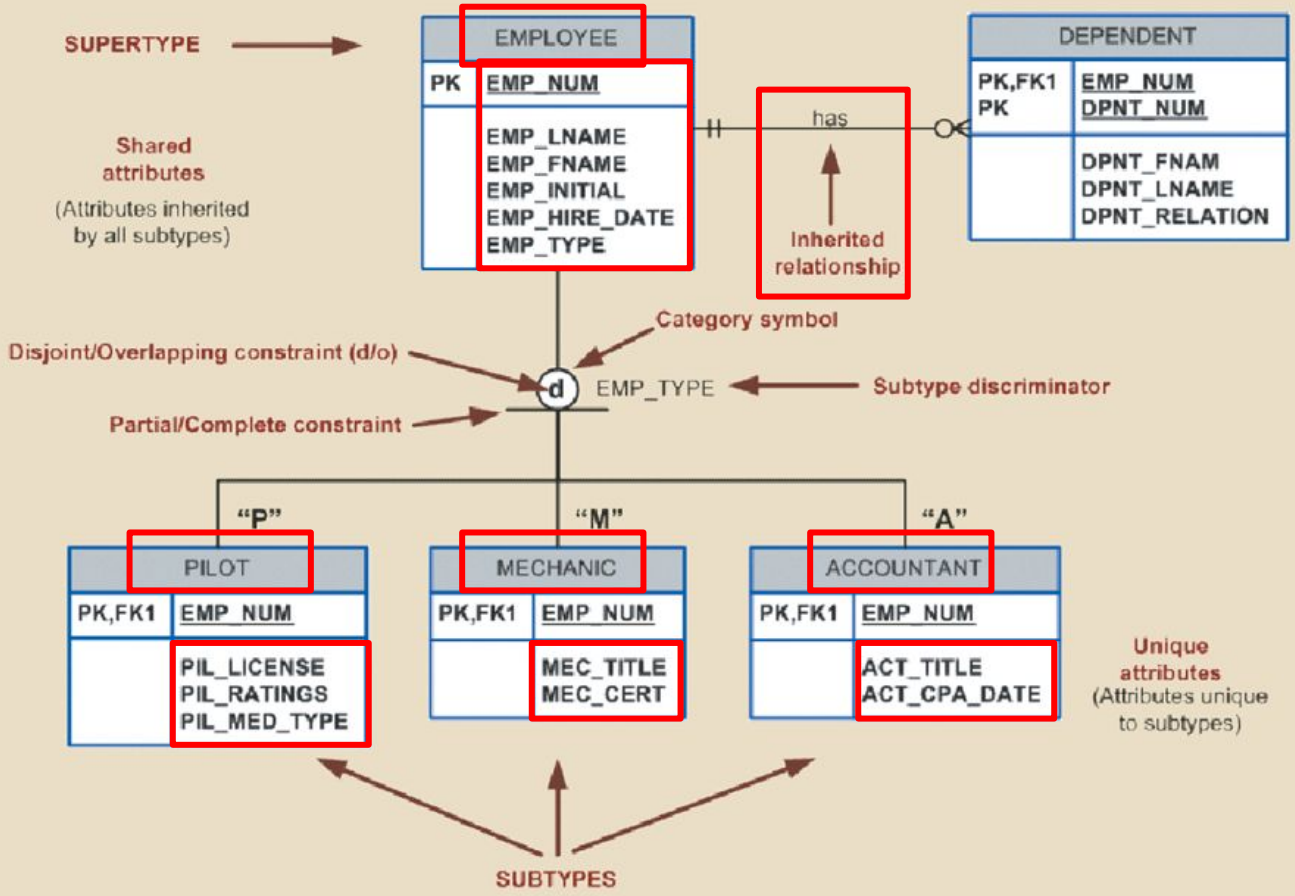


Supertype: a **generic** entity type that is related to one or more **subtypes**

Supertype contains **common characteristics**

Subtypes each contain their own **unique characteristics**

FIGURE 5.2 A SPECIALIZATION HIERARCHY



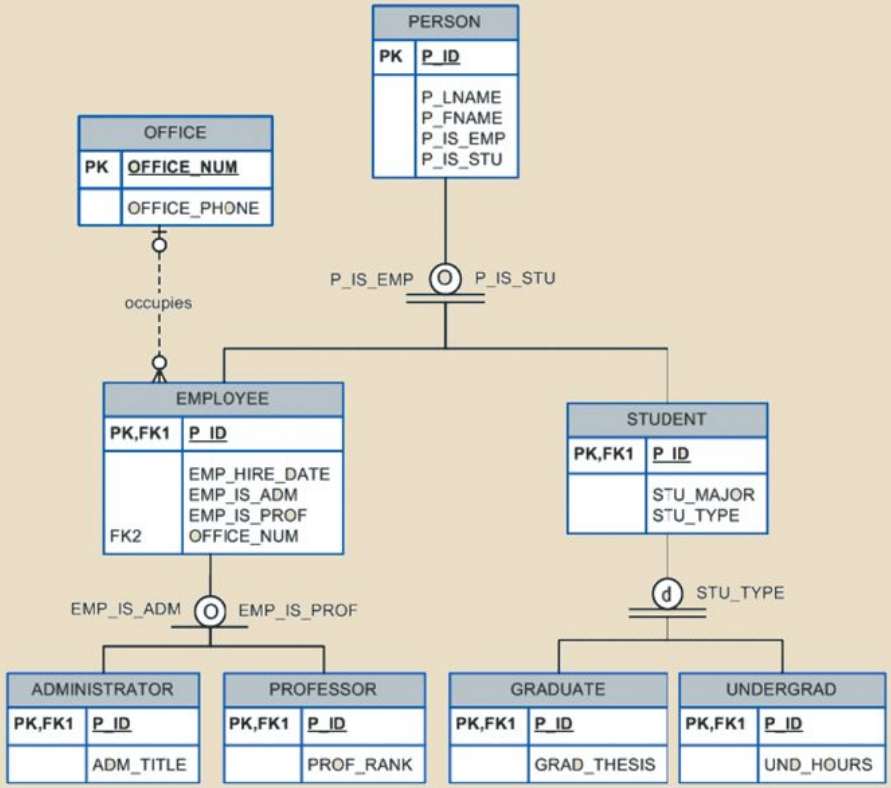
Supertype: a **generic** entity type that is related to one or more **subtypes**

Supertype contains **common characteristics**

Subtypes each contain their own **unique characteristics**

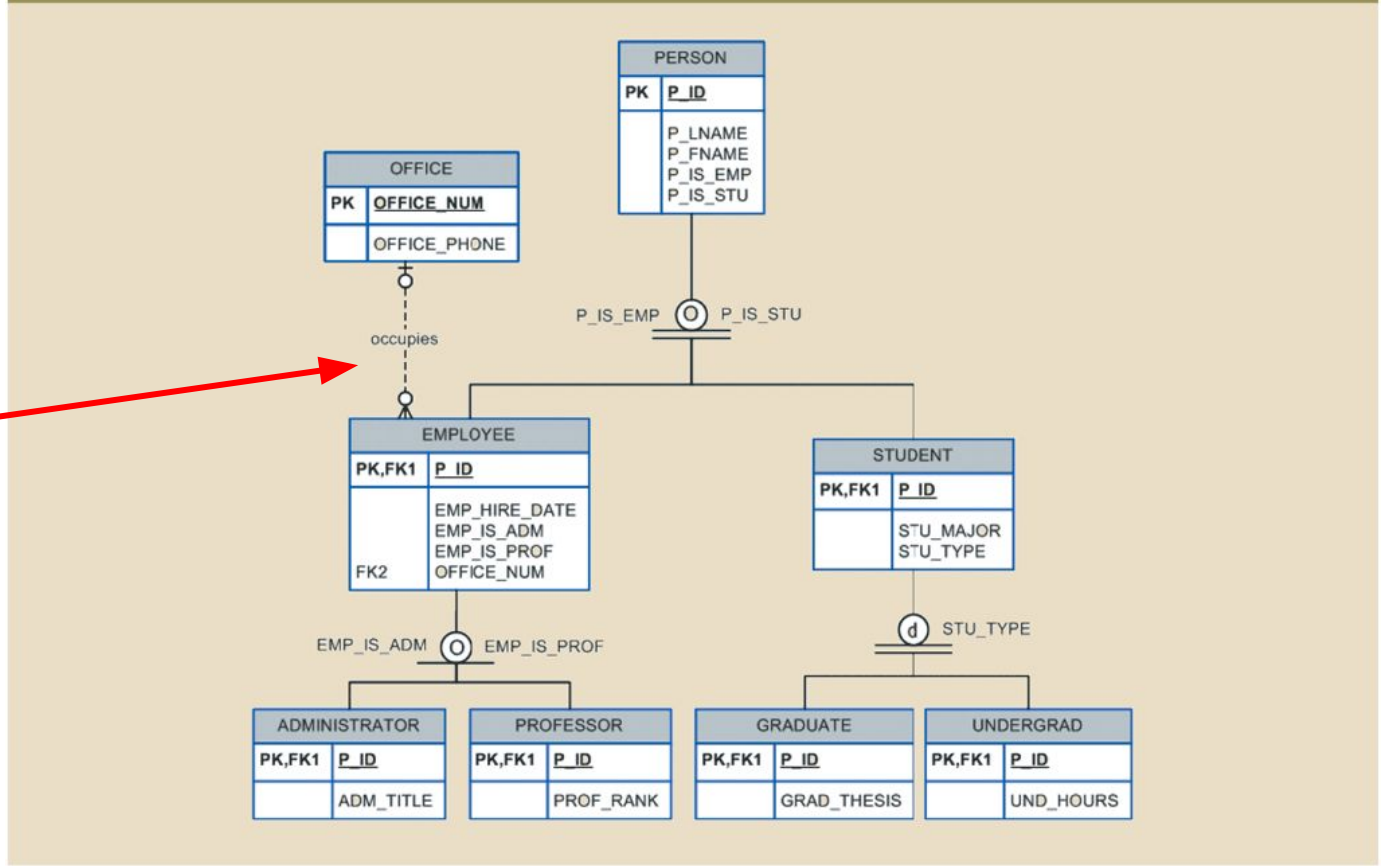
Entity subtypes **inherit all relationships** in which the supertype participates

FIGURE 5.4 SPECIALIZATION HIERARCHY WITH OVERLAPPING SUBTYPES



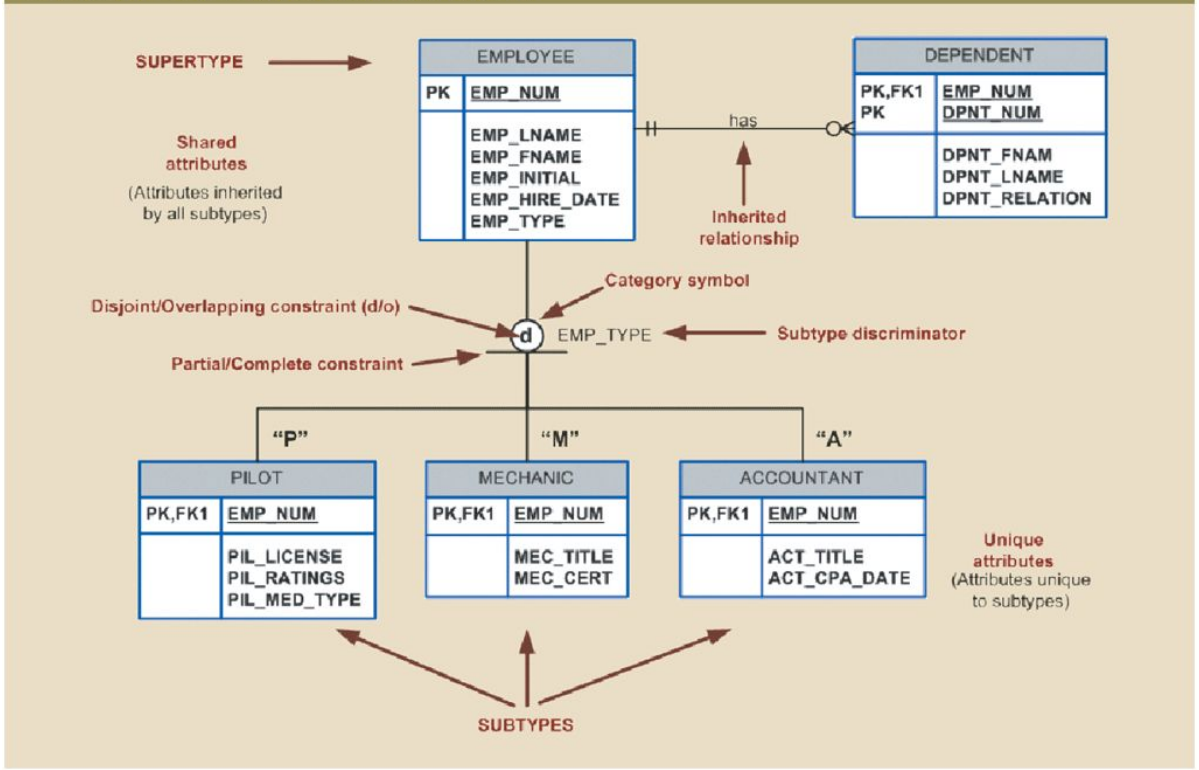
Subtypes can have **relationships of their own.**

FIGURE 5.4 SPECIALIZATION HIERARCHY WITH OVERLAPPING SUBTYPES

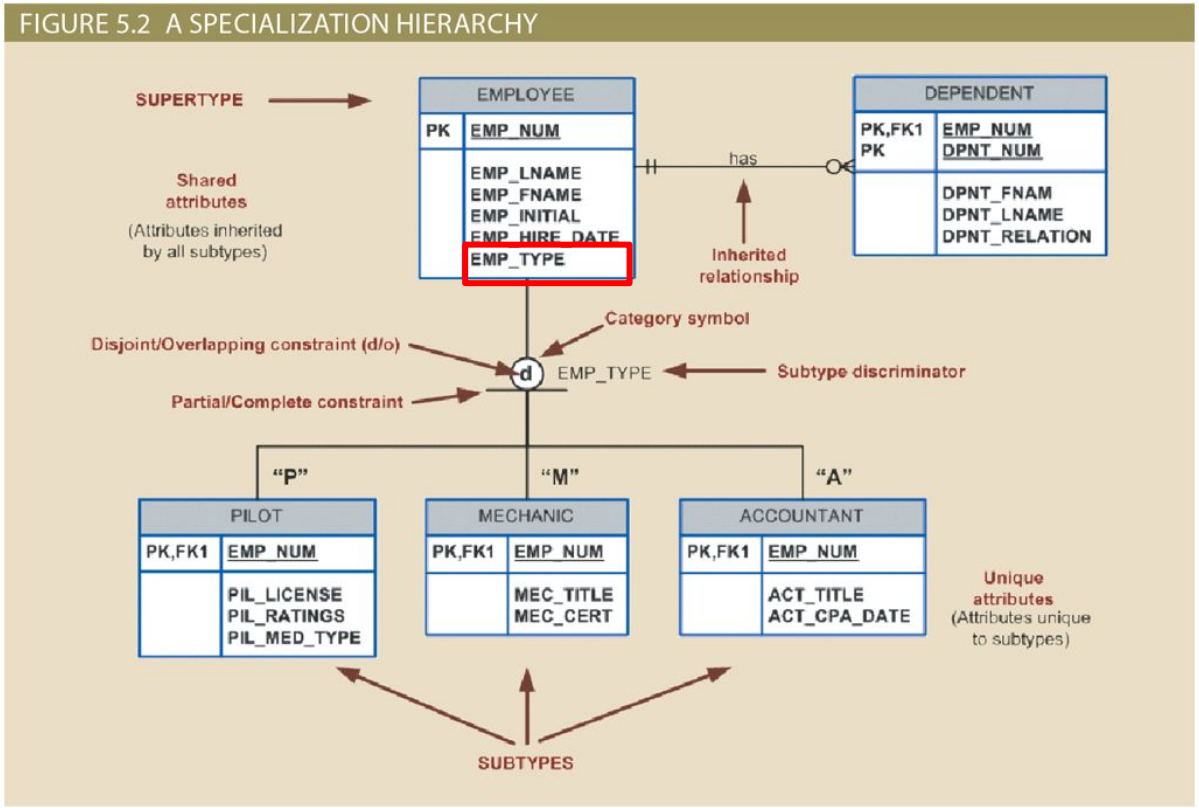


Subtype Discriminator

FIGURE 5.2 A SPECIALIZATION HIERARCHY

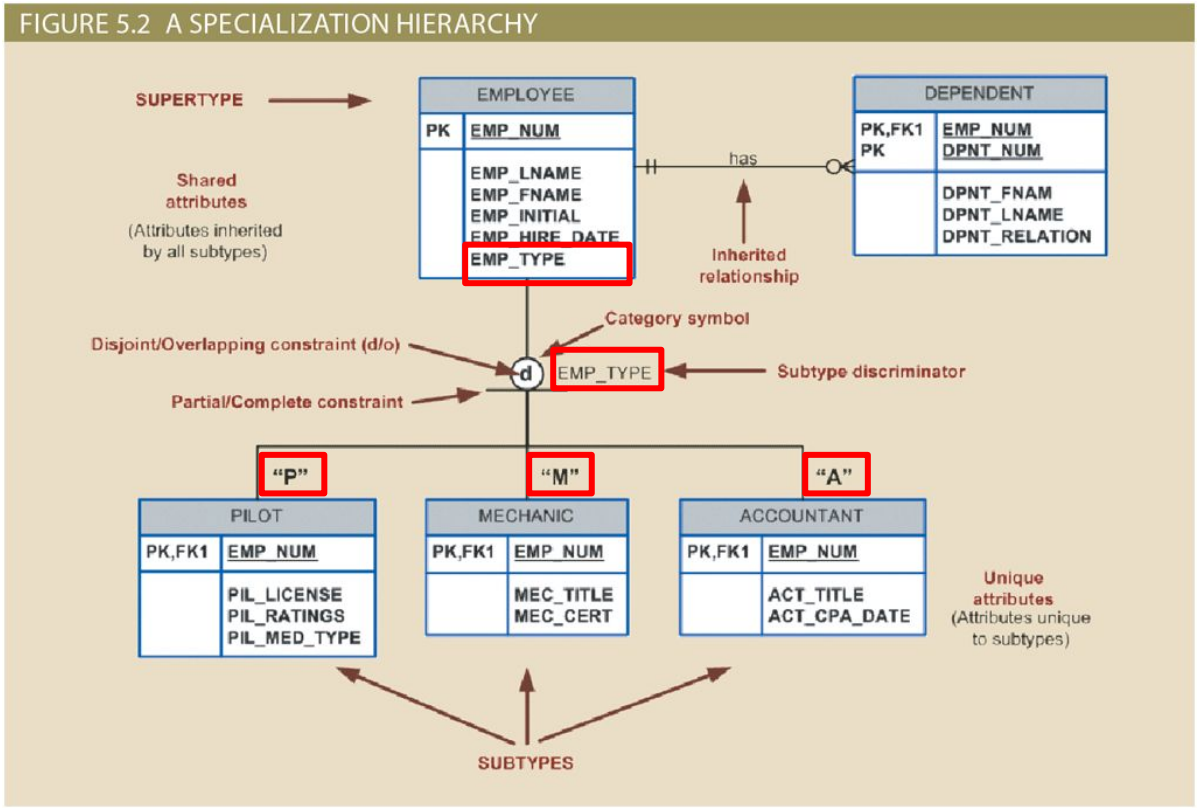


Subtype Discriminator



A property of the supertype that indicates to which subtype the supertype is related to.

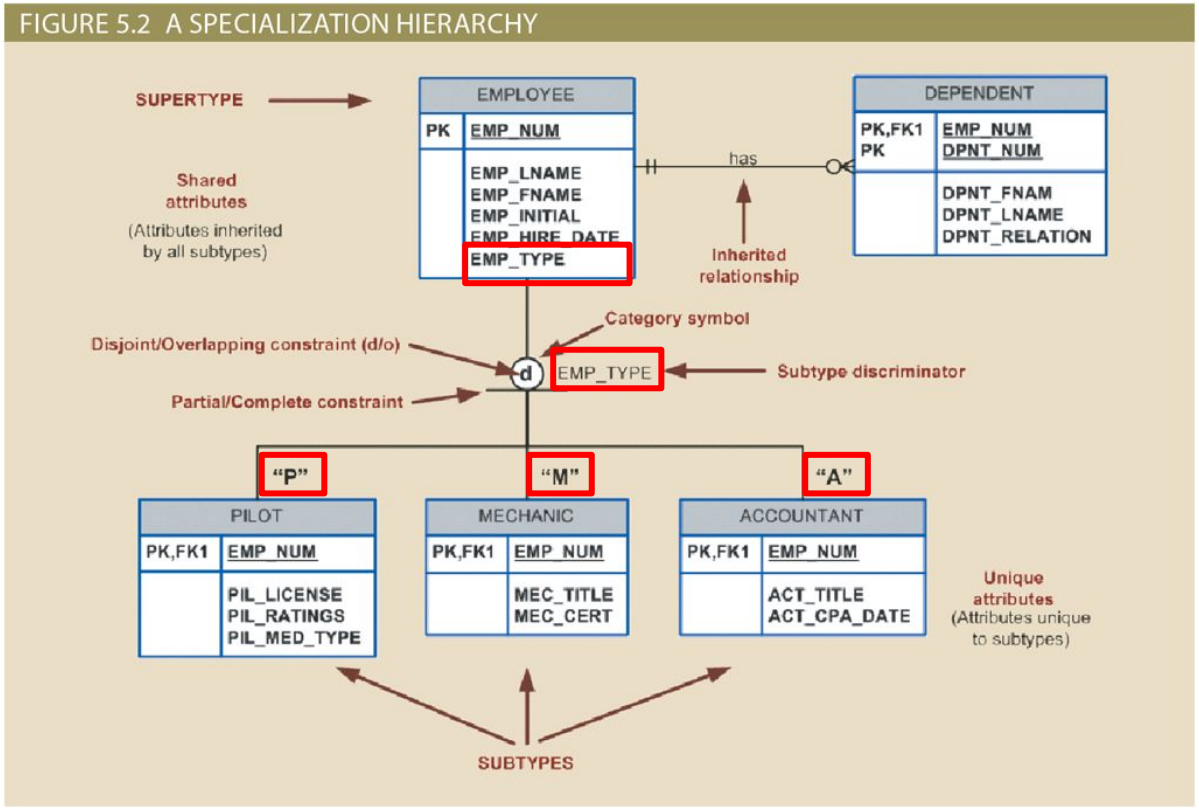
Subtype Discriminator



A property of the supertype that indicates to which subtype the supertype is related to.

It is common practice to **show the subtype discriminator and its value** for each subtype on the ERD.

Subtype Discriminator

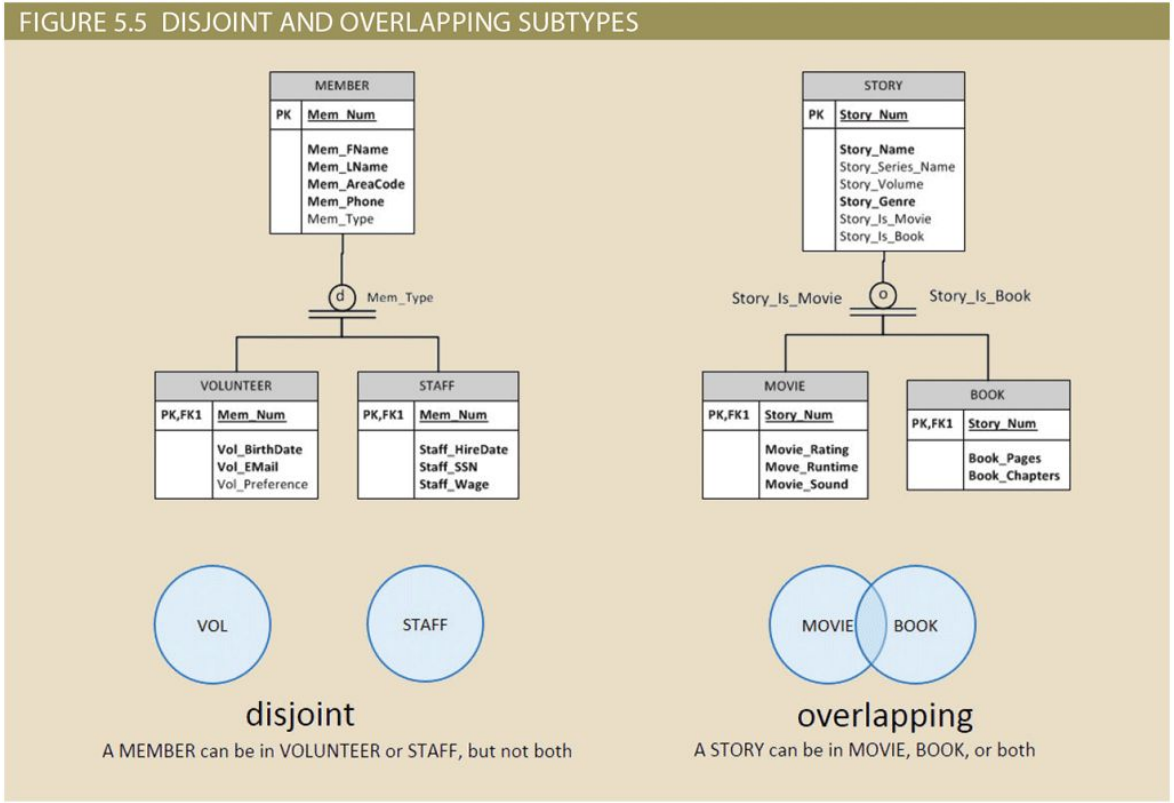


A property of the supertype that indicates to which subtype the supertype is related to.

It is common practice to **show the subtype discriminator and its value** for each subtype on the ERD.

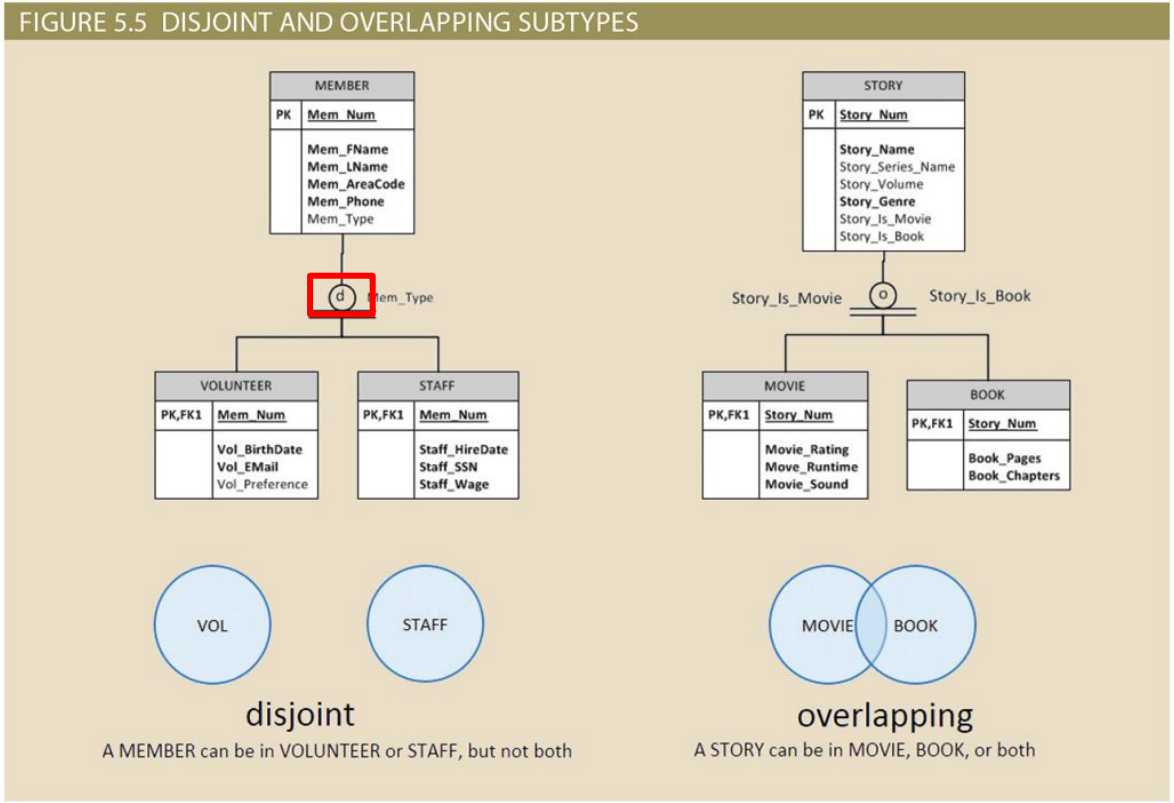
by default — but not necessarily — the subtype discriminator is based on an **equality** comparison.

Disjoint or Overlapping



The **business rules indicate** whether an entity can only be part of one subtype or more.

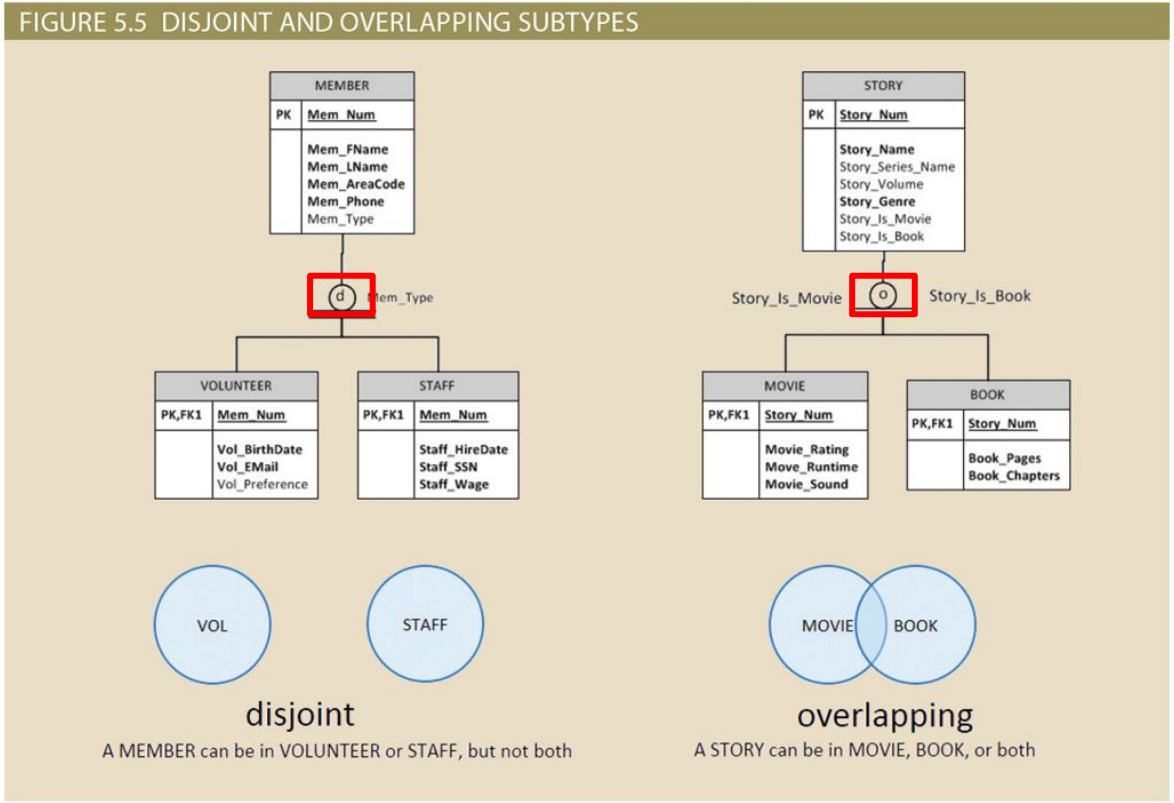
Disjoint or Overlapping



The **business rules indicate** whether an entity can only be part of one subtype or more.

Disjoint or non-overlapping subtypes are indicated with **letter d** inside the category shape.

Disjoint or Overlapping



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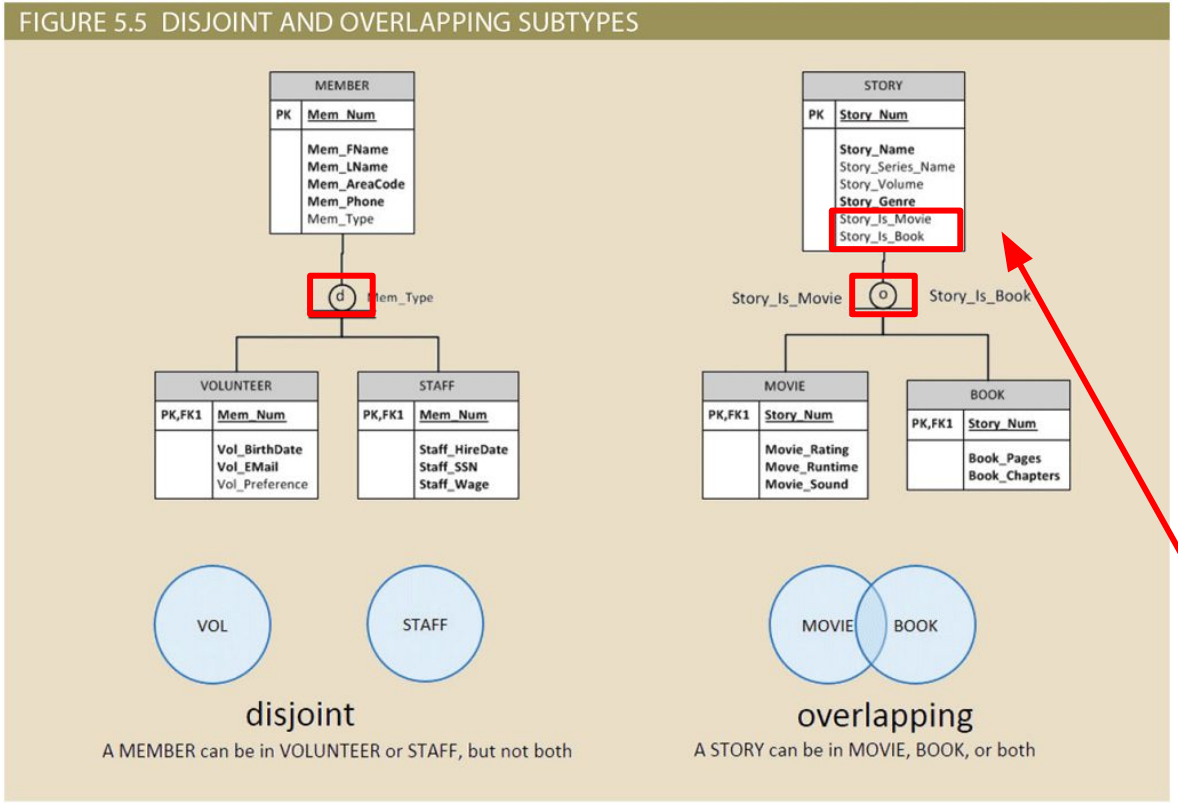
Disjoint or Overlapping



The **business rules indicate** whether an entity can only be part of one subtype or more.

Disjoint or non-overlapping subtypes are indicated with **letter d** inside the category shape.

Overlapping subtypes are indicated with the **letter o** inside the category shape

Implementing overlapping subtypes requires one discriminator attribute for each subtype.

Completeness Constraint

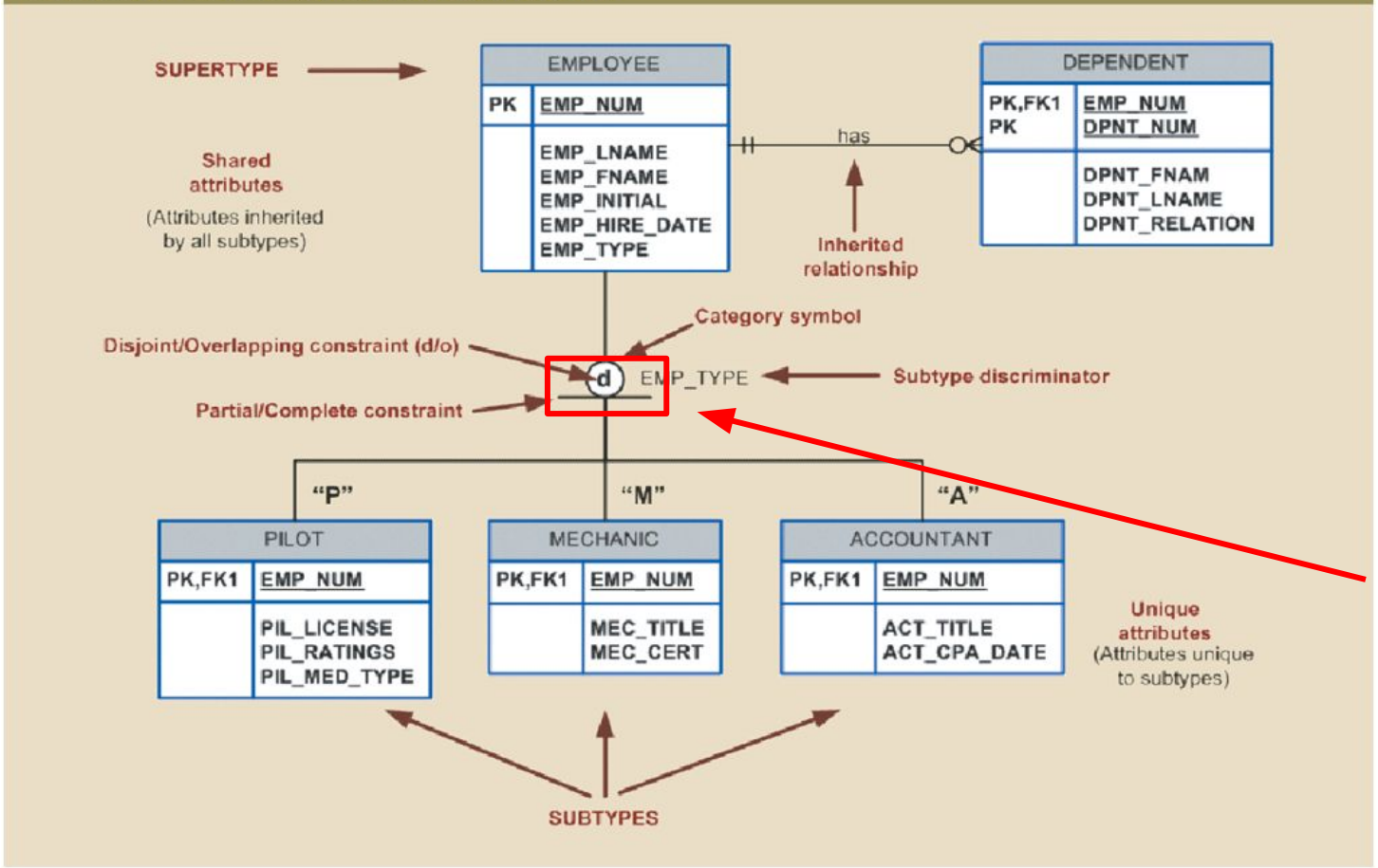
- whether each supertype occurrence is a member of at least one subtype
  - **Partial** Completeness: **not every** supertype occurrence is a member of a subtype
  - **Total** Completeness: **every** supertype occurrence must be a member of at least one subtype.
- alternative notations exists for showing partial or total completeness

TABLE 5.2

SPECIALIZATION HIERARCHY CONSTRAINT SCENARIOS

TYPE	DISJOINT CONSTRAINT	OVERLAPPING CONSTRAINT
Partial	Supertype has optional subtypes. Subtype discriminator can be null. Subtype sets are unique.	Supertype has optional subtypes. Subtype discriminators can be null. Subtype sets are not unique.
Total	Every supertype occurrence is a member of only one subtype. Subtype discriminator cannot be null. Subtype sets are unique.	Every supertype occurrence is a member of at least one subtype. Subtype discriminators cannot be null. Subtype sets are not unique.

FIGURE 5.2 A SPECIALIZATION HIERARCHY



Example

Clerk is an identifiable kind but doesn't have any attributes unique to it. So it should not be implemented as a subtype. Therefore the specialization hierarchy is modeled as **partial**.

The Extended Entity Relationship Model (EERD)

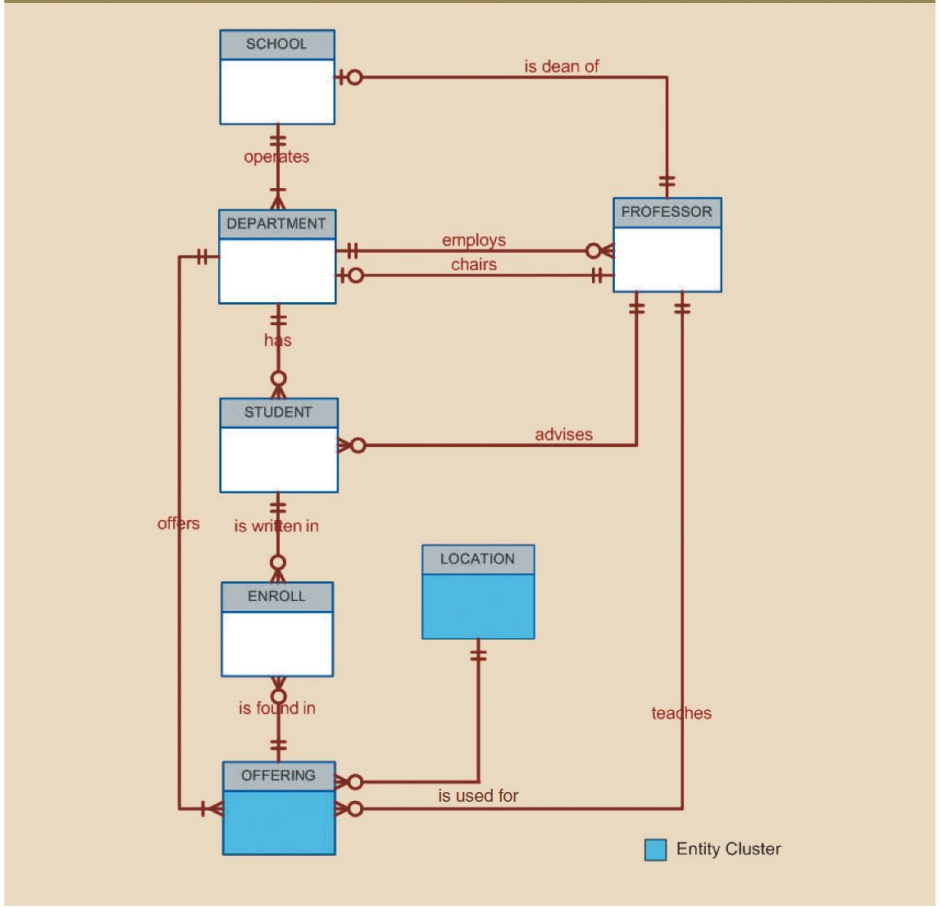
- Subtypes & Supertypes
- **Entity Clustering**

Entity Clustering

- ERDs in the real world usually include hundreds of entities and relationships which make the **ERD crowded** and not very useful as a communication tool.

- To simplifying the diagram, we can **cluster** multiple entities and the relationships between them when the entities **logically belong together**.

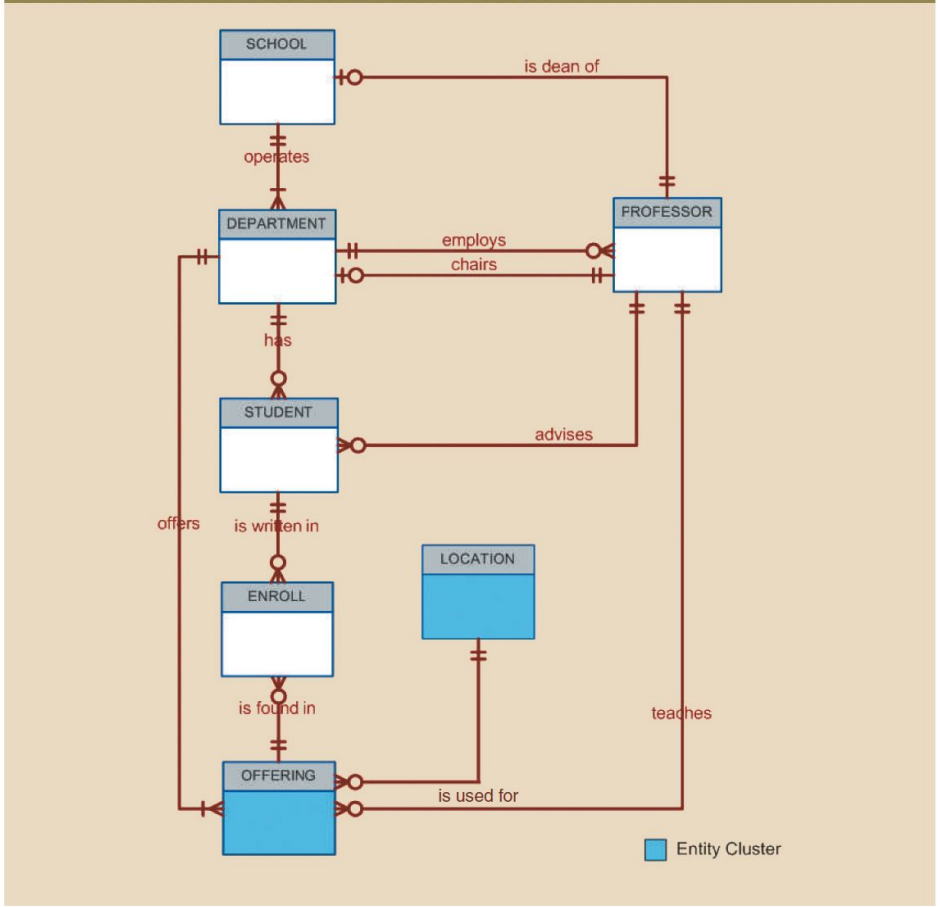
FIGURE 5.6 TINY COLLEGE ERD USING ENTITY CLUSTERS



OFFERING is a cluster which includes SEMESTER, COURSE and CLASS entities.

A LOCATION specifies both BUILDING and ROOM within the building.

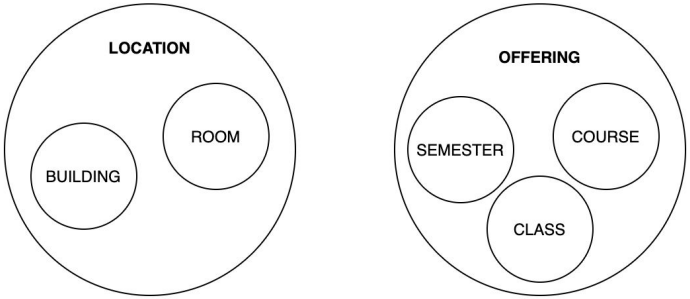
FIGURE 5.6 TINY COLLEGE ERD USING ENTITY CLUSTERS



OFFERING is a cluster which includes SEMESTER, COURSE and CLASS entities.

A LOCATION specifies both BUILDING and ROOM within the building.

In a sense we are building a **generalization hierarchy.** We are grouping these entities together which collectively represent a higher, more general concept.



Takeaways & Next Steps

By the end of this module, you should be able to:

- Recognize entities, attributes, and relationships in business rules
- Distinguish attribute types (required vs optional, composite, multivalued, derived)
- Understand relationship properties (connectivity, cardinality, strength, participation, recursive)
- Identify special entities (weak entities, associative entities)
- Apply enhanced ER concepts (supertypes, subtypes, specialization, disjoint/overlapping, completeness, clustering)

Next Steps → Module 3

- Learn how to map ER/EER diagrams into relational schemas
- See how keys, foreign keys, and constraints enforce these designs